

Inverse Kinematics Wrist Math

EGR 445/545

GVSU

Plan for Today

- schedule talk
- quick review of inverse kinematics
- inverse kinematics wrist math
- LA 09
- project time

Announcements

- HW 2 due 6/24
- Quiz 3 also 6/24
 - inverse kinematics
 - one page of notes
 - Python allowed
- LA 10 dues 6/26
- Exam 2 on 7/3
 - inverse kinematics

Quick Review of Inverse Kinematics Concepts

- inverse kinematics is hard and we often seek ways to break problems into smaller pieces
- a fully algebraic solution is tough
- we can often break an industrial inv. kin. problem in half at the wrist
- knowing how to solve simpler 3DOF problems can help with these half problems
- our approach will combined splitting problems at the wrist with a geometric approach for the first half
- we have to know the orientation of the final link to use this approach
- we need to convert the customers problem statement to ${}^0_{tip}\mathbf{T}$ to use our approach

Where is the wrist?

- two options that are often the same:
 - the base of the last link
 - where the last 2-3 $\hat{Z}$ axes intersect

Key Concepts for Inverse Kinematics

- a proper problem statement for an industrial robot should include ${}^0_{tip}\mathbf{T}$
 - what would you need to formulate ${}^0_{tip}\mathbf{T}$?
- in general, # DOF = # of joints = # parameters needed for a valid problem statement

Inverse Kinematics “Algorithm”

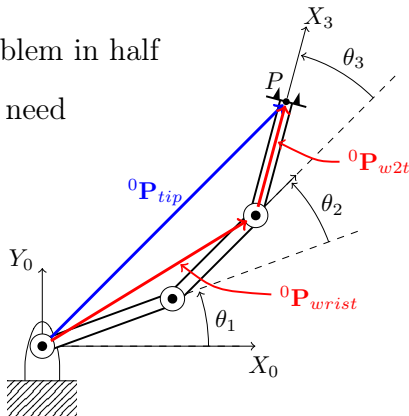
- find ${}^0P_{wrist}$
 - if we have enough information
- draw a top view
 - what variables can we solve for?
- draw a side view
 - what frame do we need to draw the side view in?
 - apply geometric approach to the side view
- verify wrist location based on side view solution
- solve for the wrist angles
- verify final answer
 - reproduce ${}^0T_{tip}$ based on forward kinematics

Finding the Wrist

- essentially splits the problem in half
- what information do we need to find the wrist?

$${}^0\mathbf{P}_{tip} = {}^0\mathbf{P}_{wrist} + {}^0\mathbf{P}_{w2t}$$

$${}^0\mathbf{P}_{wrist} = {}^0\mathbf{P}_{tip} - {}^0\mathbf{P}_{w2t}$$



$${}^0\mathbf{P}_{w2t}$$

- how do we find the wrist-to-tip vector ${}^0\mathbf{P}_{w2t}$ in frame 0?
- ${}^0\mathbf{P}_{w2t} = {}^0\mathbf{P}_{tip} - {}^0\mathbf{P}_{wrist}$
- this is the relative displacement from O_N to the tip
 - the length should be the same in any frame
- typically:

$${}^N\mathbf{P}_{w2t} = \begin{bmatrix} a_{N-1} \\ 0 \\ 0 \end{bmatrix} \text{ or } \begin{bmatrix} 0 \\ 0 \\ d_N \end{bmatrix}$$

${}^0\mathbf{P}_{w2t}$ continued

Having ${}^N\mathbf{P}_{w2t}$, how do we find ${}^0\mathbf{P}_{w2t}$?

- this is a relative displacement, so we just need to rotate it
 - we need ${}^0_N\mathbf{R}$, not ${}^0_N\mathbf{T}$
- how do we find ${}^0_N\mathbf{R}$?

From the problem statement

$${}^0_{tip}\mathbf{T} = \begin{bmatrix} {}^0_N\mathbf{R} & {}^0\mathbf{P}_{tip} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- ${}^0_N\mathbf{R} = {}^0_{tip}\mathbf{T} [0:3, 0:3]$
- ${}^0\mathbf{P}_{w2t} = {}^0_N\mathbf{R} {}^N\mathbf{P}_{w2t}$

Wrist Matrices

Problem: we are trying to find the wrist, but we don't know θ_4 yet

- the wrist is typically at O_4
 - yellow Fanuc is an example
- how can I use HT matrices to find O_4 without knowing θ_4 ?

$${}^3_4\mathbf{T} = {}^3_W\mathbf{T} {}^W_4\mathbf{T}$$

$${}^0_{tip}\mathbf{T} = {}^0_1\mathbf{T} \cdot {}^1_2\mathbf{T} \cdot {}^2_3\mathbf{T} \cdot {}^3_W\mathbf{T} \cdot {}^W_4\mathbf{T} \cdot {}^4_5\mathbf{T}$$

Wrist Matrices Continued

- ${}^3_W\mathbf{T} = {}^3_4\mathbf{T}|_{\theta_4=0} = \text{DH}(\alpha_3, a_3, 0, d_4)$
- ${}^W_4\mathbf{T} = {}^3_4\mathbf{T}|_{\alpha_3=a_3=d_4=0} = \text{DH}(0, 0, \theta_4, 0)$

Solving for Wrist Angles

Big Idea:

$${}^W_{tip}\mathbf{T}_{\text{numeric}} = {}^W_{tip}\mathbf{T}_{\text{symbolic}}$$

- Find ${}^W_{tip}\mathbf{T}_{\text{symbolic}}$ based on symbolic forward kinematics
- ${}^W_{tip}\mathbf{T}_{\text{numeric}}$ should be all hard numbers
- we have 12 equations (4x4 matrices)
 - pick options that lead to easy equations for θ_4 , θ_5 , and θ_6

Numeric Approach for ${}^W_{tip}\mathbf{T}$

- Find wrist location in frame $\{0\}$
- Find θ_1 from the top view
- Find θ_2 and θ_3 from the wrist position in frame $\{1\}$ using the geometric approach
- Once we have θ_1 , θ_2 , and θ_3 :

$${}^0_W\mathbf{T}_{\text{numeric}} = {}^0_1\mathbf{T} \cdot {}^1_2\mathbf{T} \cdot {}^2_3\mathbf{T} \cdot {}^3_W\mathbf{T}$$

$${}^0_{tip}\mathbf{T} = {}^0_W\mathbf{T} \cdot {}^W_{tip}\mathbf{T}$$

$${}^W_{tip}\mathbf{T}_{\text{numeric}} = {}^0_W\mathbf{T}^{-1} \cdot {}^0_{tip}\mathbf{T}$$

Quick Review: Finding the wrist in frame 0

$${}^0P_{wrist} = {}^0P_{tip} - {}^0P_{w2t}$$

- what code do we need to find ${}^0P_{wrist}$?

Quick Review: Side View in Frame 1

- what does this mean?
- how do I find ${}^1P_{tip}$ or ${}^1P_{wrist}$?

Fig. 3.15 applied to the wrist

$${}^3_4\mathbf{T} = {}^3_W\mathbf{T} \cdot {}^W_4\mathbf{T}$$

Solving for wrist angles

$${}^W_{tip}\mathbf{T}_{numeric} = {}^W_{tip}\mathbf{T}_{symbolic}$$

- how do we find ${}^W_{tip}\mathbf{T}_{numeric}$?
- how do we find ${}^W_{tip}\mathbf{T}_{symbolic}$?

Inverse Kin. for a “Typical” Industrial Robot

- PUMA, Fanuc, simplified PUMA, ...
- key characteristics:
 - last 2-3 $\hat{Z}$ axes intersect at the wrist
 - wrist moves in a plane set by θ_1
- approach:
 - find ${}^0P_{wrist}$
 - find θ_1 from top view of wrist location
 - find θ_2 and θ_3 from geometric approach on side view
 - probably in frame 1
 - verify wrist location: ${}^0_W\mathbf{T}$
 - find wrist angles from ${}^W_{tip}\mathbf{T}_{numeric} = {}^W_{tip}\mathbf{T}_{symbolic}$
 - verify full solution using forward kinematics

PUMA Wrist Angle Example

- `PUMA_inverse_kinematics_wrist_example.ipynb`
 - under examples on the learning activities github page

`https://github.com/ryanGT/learning\_activities\_EGR\_445\_545/blob/main/examples/PUMA\_inverse\_kinematics\_wrist\_example.ipynb`